

Computer Networks

Overview



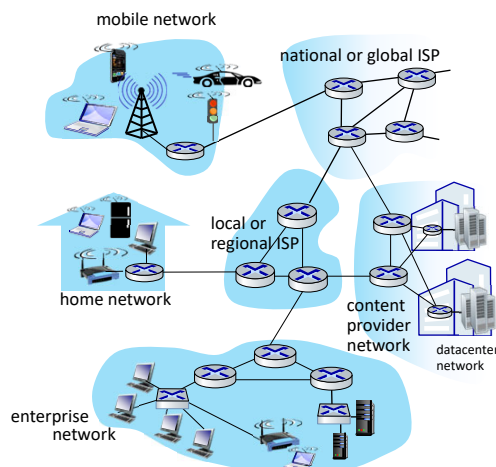
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What is a network?

A network is a collection of many diverse devices, e.g., computing and storage devices, routers, access points, connected through wired and wireless links that allow devices to communicate



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Protocols

All interactions between devices, i.e., communication activities, are governed by protocols

A *protocol* is an agreement between two parties on how communications should take place

It defines the set of rules that specify the *format* and *order* of messages exchanged (i.e., transmitted/received) between parties, and the *actions* to be performed when messages are transmitted or received

Each protocol provides some specific services that allow messages to be transferred from one end to the other end of the network

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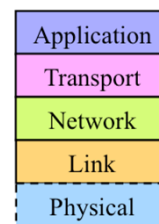
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Layered protocol stack

Protocols are organized in layers: each layer implements a specific service and relies on services provided by the layer below

Internet protocol stack consists of four (or five) layers

- Application layer protocols support communications between networked applications
- Transport layer protocols are responsible of process to process data transfer
- Network layer protocols are responsible of routing from a source to a destination
- Link layer protocols are responsible of data transfer between neighboring network elements
- *Physical layer protocols are responsible of bits transfers "on the wire"*

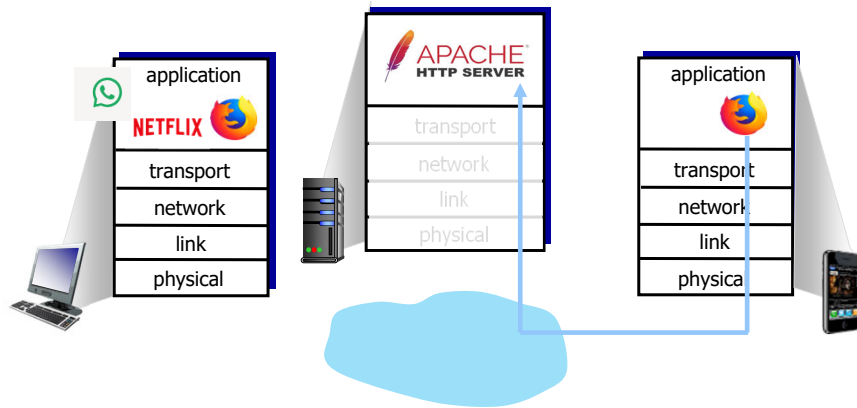


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Scenario



Two processes running on different (heterogeneous) devices communicate by exchanging *messages*

Each process is uniquely identified by the *IP address* of the device where it is running and the *port number* associated with the service being used

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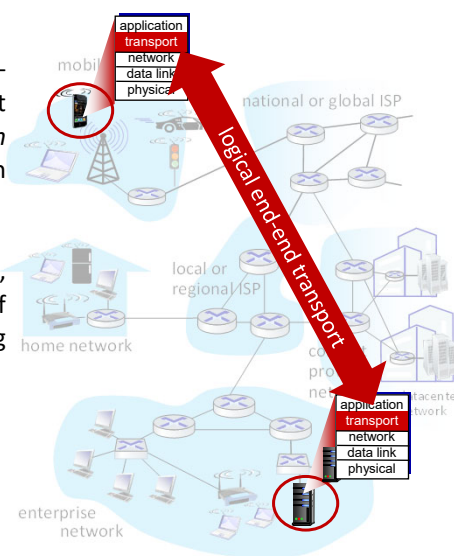
Transport layer services and protocols

Transport layer provides an end-to-end transport service, that is, it establishes a *logical communication* between processes running on different devices

Processes exchange messages, without worrying of the details of the physical infrastructure carrying the messages

Two transport layer protocols:

- Transmission Control Protocol (TCP)
- User Datagram Protocol (UDP)



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TCP protocol

Main characteristics:

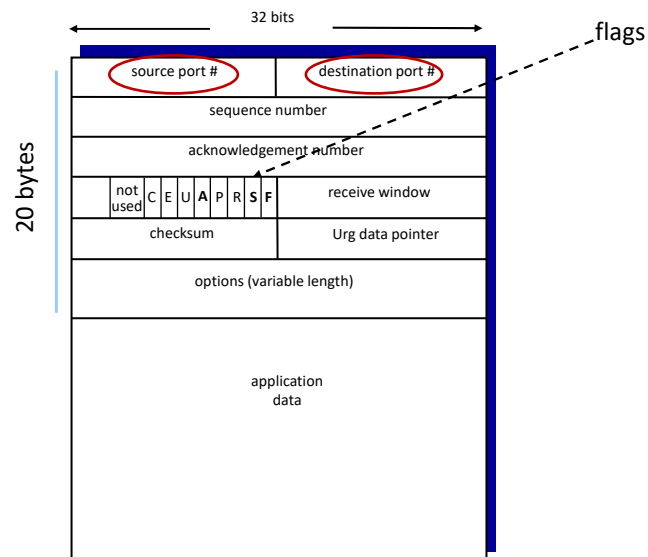
- *Connection oriented*: setup of a connection required between processes (three-way handshaking) + connection closing
- *Reliable data transfer* between sending and receiving process with in-order delivery (acknowledgement mechanisms)
- *Flow control*: sender process does not overwhelm receiver process
- *Congestion control*: throttle sender process when network overloaded

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Format of TCP packets



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UDP protocol

Main characteristics:

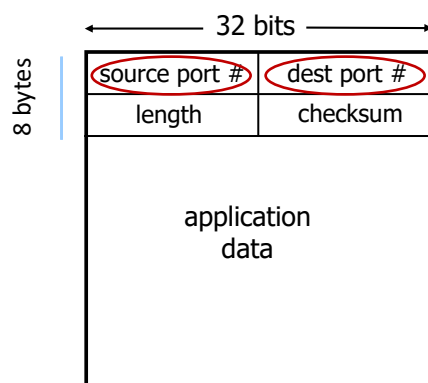
- *Connectionless*: no setup required between processes
- *Unreliable data transfer* between sending and receiving process
- *No flow control*
- *No congestion control*

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Format of UDP packets



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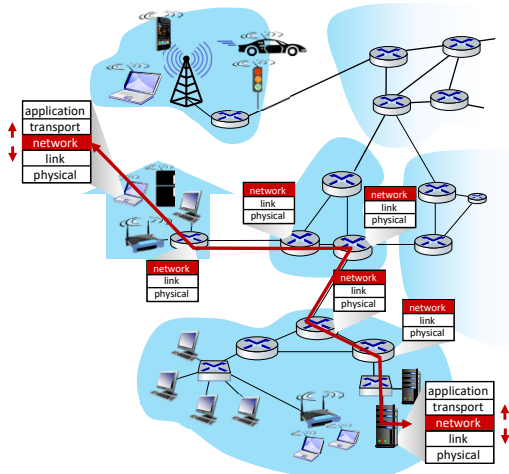
Network layer services and protocols

The main services provided by network layer protocols refer to:

- **Forwarding:** move packets from input link to appropriate output link of a router
- **Routing:** determine route taken by packets from source to destination

Two network layer protocols:

- Internet Protocol (IPv4 and IPv6)
- *Internet Control Message Protocol (ICMP)*

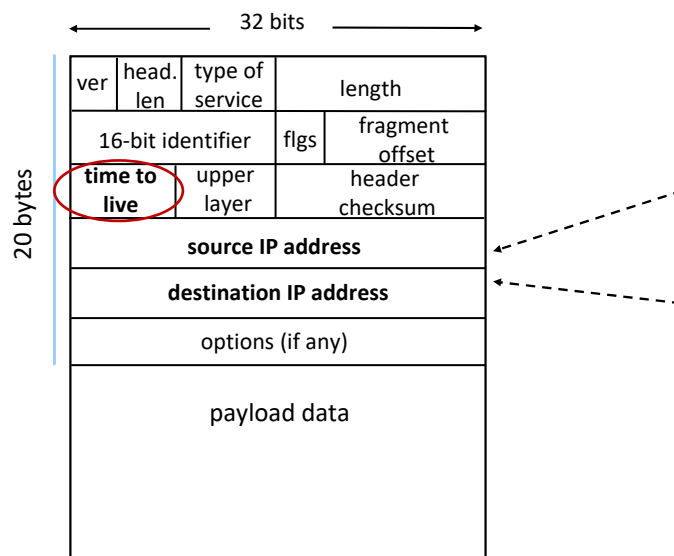


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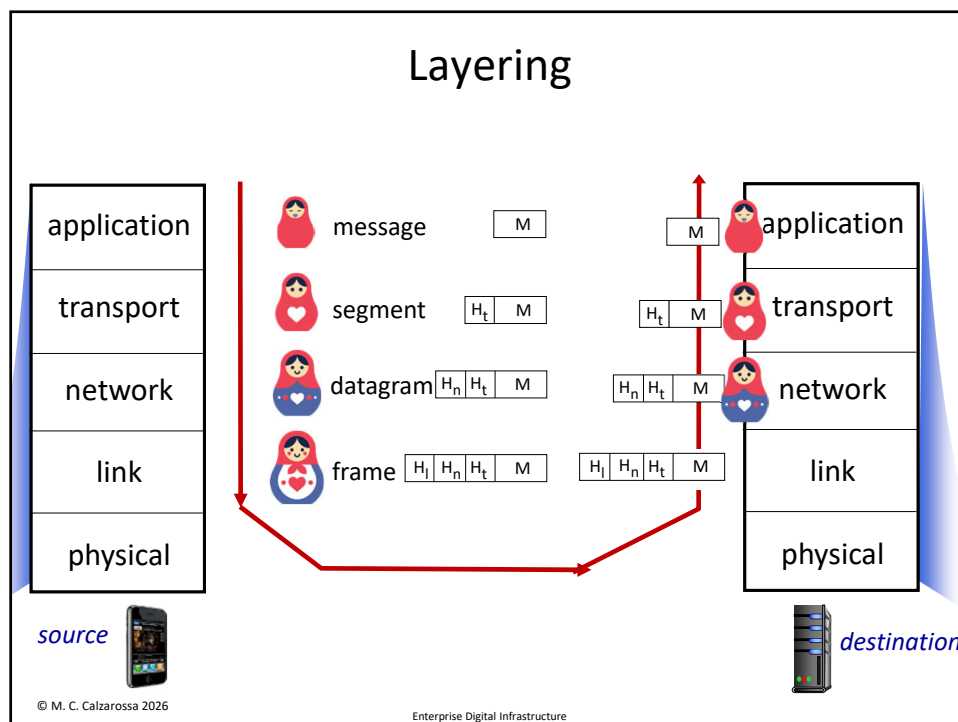
Format of IP packets



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